

NEXUS

v3.0 Sovereign Edition — Agnostic Intelligence Platform

SuperGrok Project Brief — full context for iteration

Purpose: give SuperGrok (or any coding agent) a complete, portable picture of what Nexus is, what is already built at C:\Nexus\dev, citizen-journalist utility, cross-industry use cases, workspace rules, constraints, file map, Grok Build handoffs, and next iteration targets — without dumping the repo.

Field	Value
Product	Nexus (name only — no other branding)
Version	3.0.0
Root	C:\Nexus\dev
Stack	TypeScript · React 19 · Vite 8 · Tailwind 4 · Zustand · Leaflet · R3F/Three
Audience	Citizen journalists + independent researchers (cross-industry capable)
Core job	Signal vs noise — score claims +1 / 0 / -1; export only after Layer-0
Brief date	2026-07-25 22:22 UTC
Build gate	npm.cmd run lint && npm.cmd run build (exit 0)

How to use with SuperGrok: Attach this PDF and say: “This is the current Nexus project brief. Iterate from this state. Respect KEEP publicApi, agnostic rules, and the tiled workspace plan. Repo is at C:\Nexus\dev.”

1. Contents

2. Executive summary
3. Product principles (non-negotiable)
4. What is already built (current state)
5. Architecture and file map
6. Modules (roles and interactions)
7. Evidence system, Layer-0, working document, export
8. Citizen journalism — utility and signal/noise loop
9. Cross-industry use cases (profiles, not forks)
10. Anti-clutter UX and density budget
11. Tiled workspace specification (no cascade)
12. Planned use-case catalog (IDs)
13. Procedural Forge and massing
14. Data packs and publicApi boundary
15. Grok Build handoffs and commands
16. Iteration backlog (prioritized)
17. Verification gates and evidence language
18. Explicit non-goals
19. Prompt seed for SuperGrok sessions

2. Executive summary

Nexus is a **domain-agnostic modular intelligence and modeling platform**. It is not a single-issue surveillance app and not a newsroom CMS. It is a portable **evidence workbench**: researchers load generic data packs, score material with mandatory tri-state language, place claims in space and networks, model constraints, progress through an audit ladder, and export packages only when Layer-0 clears.

The active product at C:\Nexus\dev is a runnable React shell with nine modules, Sample Pack Alpha, Layer-0 gating, working-document persistence, Procedural Forge dual export (Unity C# + Three/R3F), and an Export Kit. Legacy domain-specific UI is archived under src/legacy/ and is **out of product path**. Optional public API adapters remain under src/lib/publicApi/* and must be **KEEPed**.

Strategic north star for release: **citizen journalists and independent researchers** who need to validate claims, classify conjecture, and ship brief packages peers can audit. The same shell serves infrastructure, regulatory, network, environmental, contracting, and technical beats via use-case profiles + packs — without clutter or product forks.

Next major product step (briefed for Grok Build):

Use-case profile switcher + depth-aware **tiled locked workspace** (max 5 panes), Atlas fully visible in spatial profiles, Research + Design Lab as primaries, no cascading floating windows. Strategy: docs/USE_CASES_AND_WORKSPACE.md . Build brief: .hermes/briefs/GROK_BUILD_USECASE_WORKSPACE.md

3. Product principles (non-negotiable)

Principle	Rule
Evidence-first	Only +1 / 0 / -1. No narrative without scores.
Layer-0	Pre-filter on export, ladder L3/L4, pack replace, high-stakes actions.
Working document	Auto-log decisions, conditions, evidence, generate/rewrite, exports, Layer-0.
Explicit export	User-triggered only. Never auto-download on mount/generate.
Agnostic core	No real place, org, political, or religious branding in UI/default data.
Packs not forks	Domain depth lives in data packs + use-case profiles.
Name	Product brand is Nexus only.
publicApi KEEP	src/lib/publicApi/* preserved; do not rewrite casually.
Legacy quarantine	src/legacy/** excluded from product shell and TS include.
Density	≤5 visible panes default; progressive disclosure via ladder + on-demand modules.

Tri-state definitions

Score	Meaning	Export impact
+1	Supported — primary or strongly corroborated	Safe to carry forward
0	Insufficient or contested — hold claim	Not a soft pass
-1	Contradicted or disqualifying — escalate	Hard-blocks export while open

4. What is already built (current state)

Status: production build and lint pass on active tree (npm.cmd run build / lint → exit 0). App runs via npm.cmd run dev → http://localhost:5173.

Implemented and functional

- App shell: dark navy theme, left module switcher, header, status bar (Layer-0, WD, mode, pack)
- Command palette Ctrl/Cmd+K
- Nine modules: Information, Atlas, Design Lab, Research Hub, Analyst, Audit Ladder, Procedural Forge, Massing Viewer, Export Kit
- Core: layer0.ts, evidence.ts, workingDocument.ts + platformStore (persisted)
- Sample Pack Alpha: generic matrices, spatial nodes, graph, two asset types, sample ± evidence
- Design Lab → condition snapshot injects into Forge geometry factors
- Forge: generate / optimize / deploy hinges / before-after / Unity C# + Three TSX strings
- Export Kit: preview + download; ACK + open –1 gate
- Information module: full in-app documentation sections
- README + strategy doc + Grok Build polish/use-case briefs + launchers

Intentionally not in default shell

- Old roadside-surveillance compliance product UI (archived in src/legacy)
- Deflock/place-specific sample flows as product identity
- Floating multi-window cascade as default (planned replacement: tiled workspace)
- Auto federal API enrichment in default citizen-journo path

5. Architecture and file map

Active tree (product path)

```
src/
  App.tsx, main.tsx, index.css
  types/core.ts          – agnostic contracts
  types/audit.ts         – compat for publicApi (KEEP surface)
  core/layer0.ts | evidence.ts | workingDocument.ts
  store/platformStore.ts
  data/packs/samplePack.ts
  layout/                – (planned) tiled engine
  data/useCases/         – (planned) profile catalog
  components/
    layout/ Sidebar, StatusBar, CommandPalette
    ui/ primitives, formClasses
    modules/ Information, Atlas, DesignLab, ResearchHub,
              Analyst, AuditLadder, ProceduralForge,
              MassingViewer, MassingCanvas, ExportKit
  lib/
    forge/generators.ts
    export/exportKit.ts
    locationService.ts   – agnostic geocode helper
    publicApi/*          – KEEP
src/legacy/**           – archived; excluded from tsc
docs/USE_CASES_AND_WORKSPACE.md
.hermes/briefs/GROK_BUILD_*.md
```

Runtime split (operator machines)

Lane	Tool	Bill	Owns
Thin orchestrate	Hermes (API)	XAI_API_KEY	Plan, Layer-0, evidence, verify, briefs
Heavy code	Grok Build CLI	grok.com / SuperGrok	Multi-file implement, layout, polish
Chat iterate	SuperGrok chat	Subscription	Design dialogue using this PDF

6. Modules (roles and interactions)

Module	Role	Typical priority
Information	In-app authority docs	On-demand / first-run
Atlas	Map + network graph	Primary in spatial profiles
Design Lab	Condition / constraint matrices	Primary when testing feasibility
Research Hub	Scored notes + evidence ledger	Primary for claim work
Analyst	Command runtime + Layer-0 ops	Tertiary / triage
Audit Ladder	L0→L4 progression	Tertiary chips or research companion
Procedural Forge	Code meshes, hinges, dual export	On-demand
Massing Viewer	Live 3D	On-demand / design companion
Export Kit	Explicit packages	End of workflow / compact strip

Interaction spine: Design Lab conditions snapshot into Forge; Research/Analyst write shared evidence; Ladder records depth; Export packages WD + evidence + optional code; Atlas visualizes pack spatial/graph layers.

7. Evidence system, Layer-0, working document, export

Layer-0 high-stakes actions (representative)

- export.kit / export.unity / export.three / export.working-document
- ladder.promote.L3 / ladder.promote.L4
- datapack.replace / session.clear / forge.publish
- Export hard-blocked while unresolved –1 evidence remains

Working document auto-logs

- Session open; Layer-0 hold/clear; condition apply; evidence add/rescore
- Research notes; ladder promote/populate; forge generate/optimize/rewrite; exports

Export Kit contents

- Session header (pack, conditions, score tallies, ladder)
- Full evidence ledger with scores
- Optional working document transcript
- Optional Unity C# and Three/R3F for active asset
- Filename pattern: nexus-export-YYYY-MM-DDThh-mm-ss.md

8. Citizen journalism — utility and signal/noise loop

Release positioning: help citizen journalists and independent researchers **validate claims**, **classify conjecture**, and **identify signal through noise** with a replicable trail — not hot takes.

Canonical loop

1. Capture claim or question
2. Score material (+1 primary, 0 incomplete/contested, −1 contradicted)
3. Place on map/graph if spatial or relational
4. Apply constraints in Design Lab when “what would have to be true” matters
5. Climb Audit Ladder only as evidence supports (progressive depth)
6. ACK Layer-0 → Export Kit for peers / editors / public

Journalistic value by capability

Capability	Value
Tri-state scores	Kills vibes-as-fact; forces claim hygiene
Layer-0	Cannot quietly export unresolved contradictions
Working document	Replicable methodology trail
Research Hub	Structured notes with mandatory scores
Audit Ladder	Depth on demand — avoid overwhelm
Atlas	Ground statements in space and networks
Design Lab	Feasibility without narrative spin
Export Kit	Shareable packs only when intentional

9. Cross-industry use cases (profiles, not forks)

Rule: domain depth lives in the **pack + use-case profile**, never new top-level product brands. Same shell; switch profile; layout retile-locks.

Tier B domain families

Family	Pack themes (generic)	Atlas	Research	Design
Infrastructure / siting	Corridor nodes, clearance	Heavy	Med	Heavy
Regulatory / compliance	Control matrices, gaps	Light	Heavy	Med
Supply / distribution	Hop graphs, disclosure	Graph	Heavy	Light
Environmental / land use	Spatial + conditions	Heavy	Med	Heavy
Corporate / contracting	Award trails, entities	Med	Heavy	Light
Science / technical	Ladder + components	Light	Heavy	Forge opt.
Civic process	Timeline + sources	Light	Heavy	Light

10. Anti-clutter UX and density budget

Principle: one active profile, many available

- Use Case Switcher = single header control (not nine equal home screens)
- Profile sets default panes, layout preset, sample hints, pack binding
- Not a marketing carousel inside every module
- Sidebar always available but never duplicates floating windows

Density budget

Zone	Budget	Content
Primary	≥55% viewport	Atlas when spatial; else Research
Secondary	~25–30%	Research and/or Design Lab
Tertiary	≤15–20%	Analyst, Ladder chips, Export strip
Hidden	0 chrome	Forge, Massing, full Info — on demand

Progressive disclosure

- L0 — claim list + scores only
- L1 — add Atlas pins
- L2 — Design constraints
- L3+ — Forge/massing/systems only if profile needs them
- Export — never ambient; button + Layer-0

11. Tiled workspace specification (no cascade)

Problem to eliminate: floating windows cascade, overlap, and hide Atlas. Solution: locked tile grid with depth-aware flex.

Target layout (spatial profile)

+-----+-----+-----+			
	Research Hub	Ladder /	
ATLAS	(primary text)	Analyst	
(map+graph)		compact	
ALWAYS VISIBLE	+-----+		
	Design Lab	WD peek	
	(when active)		
+-----+-----+-----+			
Status: Layer-0 · WD · mode · pack · use-case			

Engine rules

- Default mode = tiled + locked (no free-float cascade)
- Format Layout recomputes from viewport + preset + pane weights + depth signals
- Depth weight w in $[1..5]$ from profile, evidence counts, note length, ladder level, map point count
- Mins: Atlas $\geq \sim 40\%$ width when primary; Research min $\sim 280\text{px}$; never crush map
- $\geq 1400\text{px}$ 3-column; $1100\text{--}1399$ 2-column; <1100 tabbed primary (still no cascade)
- Max 5 visible panes; open module focuses or replaces lowest unpinned tertiary
- Pinned Atlas never evicted in spatial profiles
- Resize via splitters; persist per use-case id; double-click reset
- Maximize/focus allowed; cascade forbidden in default mode

Planned implementation files

- `src/layout/{types,measure,presets,depthWeights,formatLayout}.ts`
- `src/components/layout/{Workspace,PaneFrame,SplitTree}.tsx`
- `src/data/useCases/catalog.ts` + store `setUseCase()`
- App.tsx: Sidebar | Header+UseCaseSwitcher+FormatLayout | Workspace | StatusBar

12. Planned use-case catalog (IDs)

ID	Label	Layout	Default open panes
cj-claim-triage	Claim triage desk	triage-compact	Research, Ladder, Analyst
cj-site-verify	Site / scene verify	spatial-primary	Atlas (pin), Research, Design
cj-network-map	Actor / flow map	spatial-primary	Atlas, Research, Analyst
cj-public-record	Public record trail	research-first	Research, Ladder, Export compact
cj-constraint-test	Constraint test	design-primary	Design, Atlas/Massing, Research
cj-package-brief	Publishable brief	export-review	Research, Ladder, Export, Analyst
gen-explore	General explore	research-first / mild spatial	Atlas + Research + Design

Generic sample claims for triage (examples): +1 public agenda document; 0 vague density claim without count; -1 claim contradicts primary timestamp.

13. Procedural Forge and massing

- Pure code hierarchical meshes; hinge deploy drivers 0→1
- Workflow: Generate → Optimize → Animate → Compare → Export
- Dual output: Unity C# generator string + Three.js/R3F component string
- Active Design Lab conditions inject into dimensions (height, arm, setback, scale)
- Every rewrite logged to working document
- Asset types in sample pack: Mast Enclosure A, Cabinet Node B
- Disclaimer: model from structured inputs — not certified survey/engineering drawing

14. Data packs and publicApi boundary

Data pack

DataPack interface (core types): meta, conditionMatrices, spatialPoints, graphNodes/Edges, assetTypes, sampleEvidence, sampleSources. Sample Pack Alpha is fully generic (Jurisdiction 01, Device Type A, Node Alpha, Condition Matrix Alpha).

How to add a pack

- Implement DataPack under src/data/packs/
- Generic labels only
- loadDataPack() requires Layer-0 ACK
- Optional: register Forge asset builders in generators.ts buildParts()

publicApi KEEP boundary

src/lib/publicApi/* (usaspending, censusGeo, openFema, evidenceBridge, enrich*, rateLimit, types) is preserved for optional enrichment. Vite proxies in vite.config.ts. Agnostic shell must not hard-bind product identity to these adapters. Citizen-journalist default path does not require live federal API calls.

15. Grok Build handoffs and commands

Task	Brief path	Launcher
Polish v3 shell	.hermes/briefs/GROK_BUILD_POLISH.md	RUN_GROK_BUILD_POLISH.cmd
Use cases + tiled workspace	.hermes/briefs/GROK_BUILD_USECASE_WORKSPACE.md	RUN_GROK_BUILD_USECASE_WORKSPACE.cmd
Strategy (read)	docs/USE_CASES_AND_WORKSPACE.md	—

Primary command (use-case workspace)

```
cd /d C:\Nexus\dev && set PATH=%USERPROFILE%\grok\bin;%PATH% && grok -p --prompt-file
"C:\Nexus\dev\.hermes\brieves\GROK_BUILD_USECASE_WORKSPACE.md" --always-approve --max-turns 50 --output-format
plain --cwd "C:\Nexus\dev"
```

Prerequisites: grok login (grok.com pool), npm.cmd install in repo.

After Build always verify

```
npm.cmd run lint && npm.cmd run build && npm.cmd run dev
```

Smoke: cj-site-verify (Atlas visible, no overlap) · cj-claim-triage (Research-led, ≤5 panes) · export blocked on open -1 · ACK then download.

16. Iteration backlog (prioritized)

P0 — Ship workspace + use cases

- Implement use-case catalog + switcher + setUseCase
- Tiled Workspace engine + PaneFrame + depth weights
- Wire App; Format Layout; persist layout per profile
- Information section for independent researchers

- Lint/build clean; agnostic scrub

P1 — Citizen journalist readiness

- Richer generic sample claims/sources for triage profile
- Compact Export strip + full Export maximize
- Empty states that teach the signal/noise loop in one screen
- Optional print-friendly export CSS
- Keyboard path completeness (pane focus cycle)

P2 — Performance and packaging

- Dynamic import code-split for Three/Leaflet if low risk
- Desktop launcher polish (existing start-nexus.cmd / launch-nexus.ps1)
- Share zip script → releases if desired
- E2E smoke (Playwright) for use-case switch + layout

P3 — Optional enrichment (not identity)

- Opt-in publicApi enrich behind explicit Research actions
- New packs for generic infrastructure/regulatory demos
- Do not rebrand product around any single domain

17. Verification gates and evidence language

All status claims must use tri-state evidence: **+1** verified by command/output, **0** incomplete/untested, **-1** failed or contradicted. Escalate any -1.

Gate	Command / check	Pass
Typecheck + build	npm.cmd run build	exit 0
Lint active tree	npm.cmd run lint	exit 0, no active warnings
Agnostic scrub	rg forbidden brands in active UI/pack/README	no hits
publicApi present	dir src\lib\publicApi	files intact
Legacy excluded	tsconfig exclude src/legacy	not imported by App
Layout integrity	spatial profile smoke	Atlas visible, no cascade
Export discipline	open -1 then export	blocked until resolve/ACK rules

18. Explicit non-goals

- Nine simultaneous equal windows or cascade stacks as default
- Domain-specific product renames per beat
- Auto-opening every module on pack load
- Auto-download exports
- Hard-coded real jurisdictions as product identity
- Turning Research into a social feed
- Rebuilding legacy roadside compliance UI into the shell
- Auth/paywall/social as blockers to local research use
- Burning Hermes API via delegate_task for multi-file coding (use Grok Build CLI)

19. Prompt seed for SuperGrok sessions

Copy-paste opener when attaching this PDF:

You are helping iterate Nexus v3 at C:\Nexus\dev. Read this project brief PDF as source of truth for goals and constraints. Nexus is an agnostic evidence workbench for citizen journalists and cross-industry research: tri-state +1/0/-1, Layer-0, working document, explicit export, data packs, nine modules. KEEP src/lib/publicApi/*. Do not reattach src/legacy into the shell. Next build: use-case profiles + depth-aware tiled workspace (max 5 panes), Atlas fully visible in spatial profiles, Research + Design primary, no cascading windows. Brief: .hermes/briefs/GROK_BUILD_USECASE_WORKSPACE.md. Heavy multi-file coding → Grok Build CLI with --prompt-file. Verify with npm.cmd run lint and npm.cmd run build. Stay agnostic: generic labels only. Propose the smallest next diff that increases citizen-journalist signal/noise utility.

End of brief. Repo path: C:\Nexus\dev · Product: Nexus · Version: 3.0.0 · Strategy: docs/USE_CASES_AND_WORKSPACE.md · This file: docs/Nexus_v3_SuperGrok_Project_Brief.pdf

America First | Truth-Seeking — evidence over narrative; portable over branded; explicit export only.